

Credit Card Data Analysis Report

DS3003 Data Processing Workshop II - 2026 Spring Project

Item	Content
Project option	Credit card data analysis
Report file name	15_NikoNikoNi_Credit_card_data_analysis_Report.pdf
Group No.	15
Group Name	NikoNikoNi
Notebook / engineering sources	credit_overdue_analysis_appended_with_ml_english.ipynb; cs-training.csv; UCI_Credit_Card.csv; credit_card/*.png

This revised report summarizes the full credit risk workflow in the engineering files: reproducing the original credit overdue analysis, repeating the analysis on the UCI credit card default dataset, and extending the project with classification and clustering models.

Revision note: group information and member contribution were updated from the final PPT. Figures and numerical results were cross-checked against the engineering ZIP outputs.

Team Contribution

Member	Main responsibility from PPT and project files	Contribution
Wang Jinxi (2430026192)	Credit project reproduction; data cleaning; Spark descriptive analysis; part of report writing.	34%
Hao Jinshuo (2430026056)	UCI credit dataset analysis; credit ML and clustering; visualization explanation.	33%
Wang Qiangchun (2430026197)	Movie filtering and movie ML; final report integration; presentation preparation.	33%

Executive Summary

The credit card project studies overdue and default risk using two datasets. It first reproduces the original credit overdue workflow and then expands the analysis to the UCI credit card default dataset. The machine learning extension compares classification models and uses K-Means to segment customers into risk profiles.

- The original credit overdue dataset was cleaned from 150,000 raw rows to 149,391 records, with an overall overdue rate of 6.70%.
- Historical overdue behavior was the strongest descriptive signal: customers with both 30-59 day and 60-89 day overdue records reached a 43.60% overdue rate.
- The UCI dataset contains 30,000 records and shows a 22.12% overall default rate.
- Credit limit and repayment delay status are strongly related to default risk; PAY_0 is the clearest repayment signal.
- Random Forest outperformed Logistic Regression, with AUC-ROC = 0.7710, accuracy = 0.8130, and F1 = 0.7897.
- However, the Random Forest default-class recall is only 33.89%, showing that accuracy alone is misleading for credit risk.
- K-Means selected K = 5 and identified Cluster 1 and Cluster 3 as the main high-risk customer groups.

Project Requirement Mapping

Requirement	How this report addresses it
Follow the document and implement everything	Step 1 reproduces the original credit overdue analysis with Pandas, Spark, and Matplotlib.
Use another credit risk dataset and redo the analysis	Step 2 repeats cleaning, Spark aggregation, and visualization using the UCI credit card default dataset.
Apply regression/classification/clustering methods	Step 3 applies Logistic Regression, Random Forest classification, and K-Means clustering.
Describe interesting results	The report interprets repayment history, credit limit, model recall, feature importance, and customer segments.
Include important code and screenshots	Selected code snippets and key figures from the engineering ZIP outputs are included with captions.

1. Introduction

Credit risk analysis helps financial institutions identify customers who may fail to repay credit card debt. This project uses both descriptive analysis and predictive modeling to study default risk. The same pipeline is used across the datasets: clean the data, aggregate with Spark, visualize key risk factors, train models, and interpret business meaning.

1.1 Tools and Libraries

Tool / Library	Purpose
Pandas	Read CSV files, inspect missing values, remove duplicates, and save cleaned datasets.
PySpark / Spark SQL	Create Spark sessions, load cleaned data, and calculate scalable group-by risk statistics.
Matplotlib	Generate overdue/default charts, comparison charts, confusion matrix, and clustering visuals.
PySpark MLlib	Build Logistic Regression, Random Forest, and K-Means models.

1.2 Overall Workflow

1. Read raw CSV data with pandas
2. Remove duplicates and handle missing values
3. Save a cleaned dataset for Spark
4. Load the cleaned data into a Spark DataFrame
5. Use groupBy and aggregation to calculate risk statistics
6. Visualize the results with matplotlib
7. Train classification models and compare metrics
8. Apply K-Means clustering to identify customer segments
9. Interpret model limits and business implications

2. Step 1 - Reproducing the Original Credit Overdue Analysis

The first part follows the original credit overdue analysis workflow using cs-training.csv. After preprocessing, the cleaned dataset contains 149,391 records and 7 selected variables.

2.1 Data Preprocessing

```
import pandas as pd
RAW_CSV = "cs-training.csv"
CLEAN_CSV = "data.csv"
df = pd.read_csv(RAW_CSV, index_col=0)
df = df.drop_duplicates()

for col in ["MonthlyIncome", "NumberOfDependents"]:
    df[col] = df[col].fillna(df[col].mean())

df_clean = df.rename(columns={
    "SeriousDlqin2yrs": "y",
    "NumberOfTime30-59DaysPastDueNotWorse": "days_30_59",
    "NumberRealEstateLoansOrLines": "RealEstateLoans",
    "NumberOfTime60-89DaysPastDueNotWorse": "days_60_89",
    "NumberOfDependents": "families"
})["y", "age", "days_30_59", "MonthlyIncome", "RealEstateLoans", "days_60_89", "families"]
df_clean.to_csv(CLEAN_CSV, index=False)
```

Item	Result
Raw data shape	150,000 rows x 11 columns
After removing duplicates	149,391 rows
MonthlyIncome missing values	29,221 values, filled with mean 6,675.10
NumberOfDependents missing values	3,828 values, filled with mean 0.76
Cleaned data shape	149,391 rows x 7 columns

2.2 Spark Statistical Analysis

```
from pyspark.sql import SparkSession, functions as F
spark = SparkSession.builder.appName("CreditOverdueAnalysis").master("local[*]").getOrCreate()
sdf = spark.read.csv("data.csv", header=True, inferSchema=True)

age_expr = (
    F.when(sdf["age"] <= 30, "0-30")
    .when(sdf["age"] <= 45, "30-45")
    .when(sdf["age"] <= 60, "45-60")
    .when(sdf["age"] <= 75, "60-75")
    .otherwise("75-100")
)
age_stats = sdf.withColumn("age_group", age_expr).groupBy("age_group", "y").count()
```

Risk factor	Observed result
Overall overdue rate	6.70% (10,009 overdue cases out of 149,391 records)
Age	Younger customers have higher overdue risk; the 0-30 group has an overdue rate of 11.71%.
Historical overdue records	Customers with both 30-59 day and 60-89 day records have a 43.60% overdue rate.
Real estate loans	Customers without real estate loans have a higher overdue rate (8.38%) than those with loans (5.71%).
Dependents	Customers with dependents show a slightly higher overdue rate (7.83%) than those without dependents (5.88%).
Monthly income	Customers below mean income have a higher overdue rate (8.16%) than those above mean income (5.18%).

2.3 Visual Results and Interpretation

Overall Credit Card Overdue Distribution

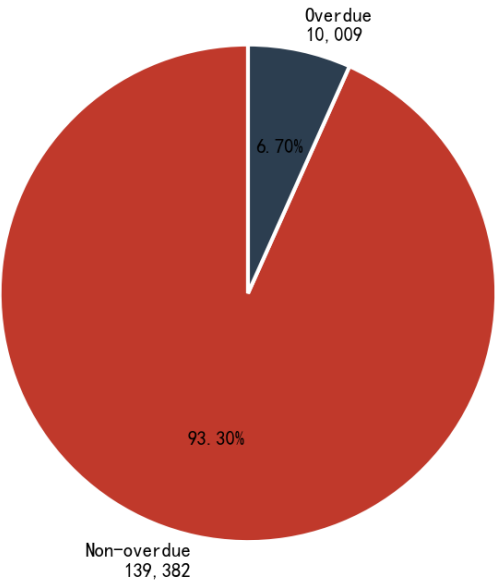
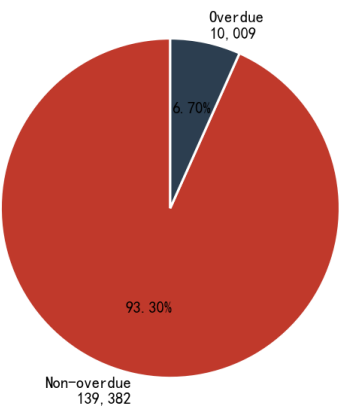


Figure 1.1. Overall overdue distribution.

Age and Credit Card Overdue Analysis

Total Overdue Distribution by Age Group



Overdue Status by Age Group (Stacked)

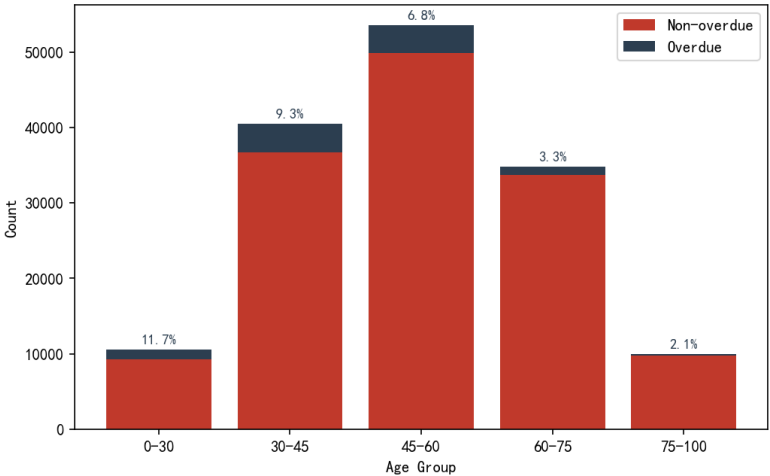


Figure 1.2. Overdue rate by age group.

Historical Overdue Records and Current Credit Card Overdue Analysis

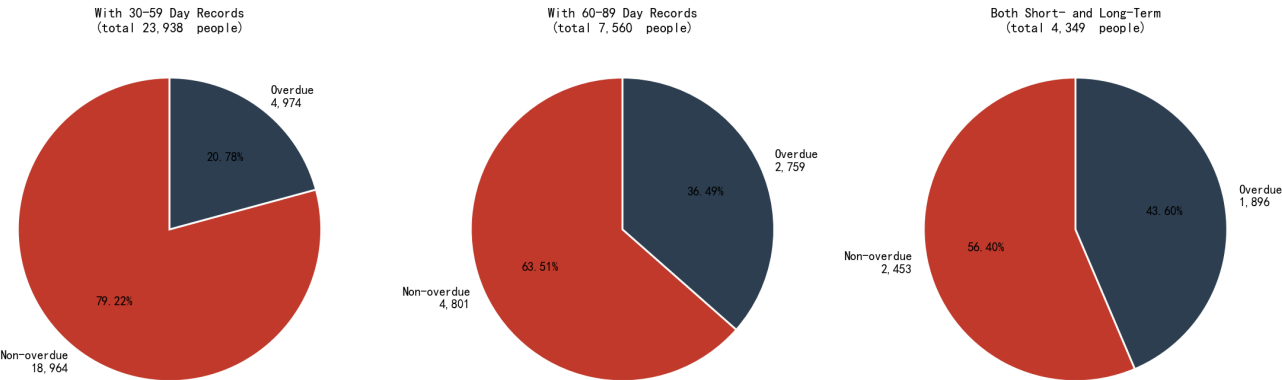
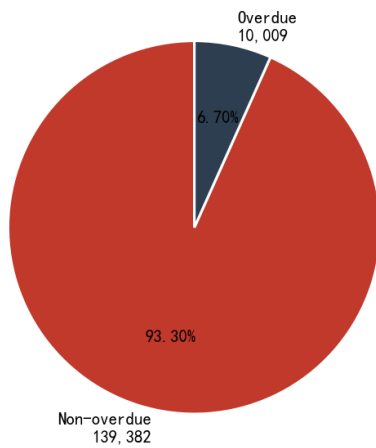


Figure 1.3. Historical overdue records and current overdue risk.

Real Estate Mortgage and Credit Card Overdue Analysis

Total Overdue Distribution by Real Estate Mortgage Status



Real Estate Mortgage and Overdue Status

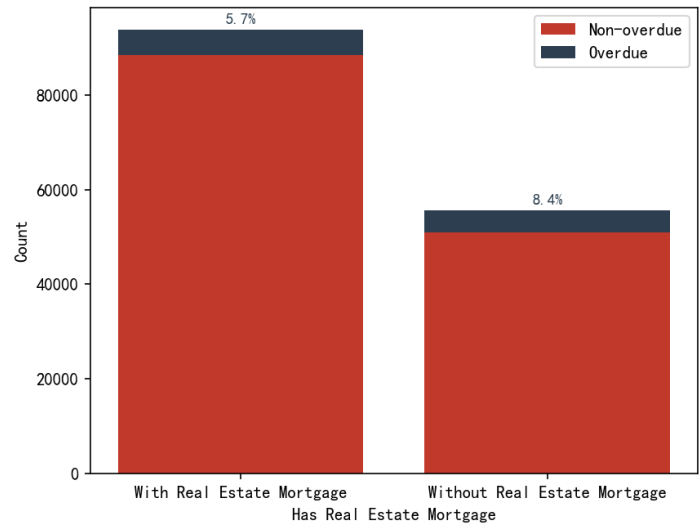
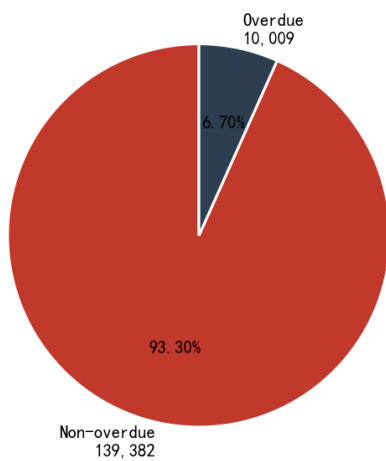


Figure 1.4. Real estate loans and overdue risk.

Number of Dependents and Credit Card Overdue Analysis

Total Overdue Distribution by Number of Dependents



Number of Dependents and Overdue Status

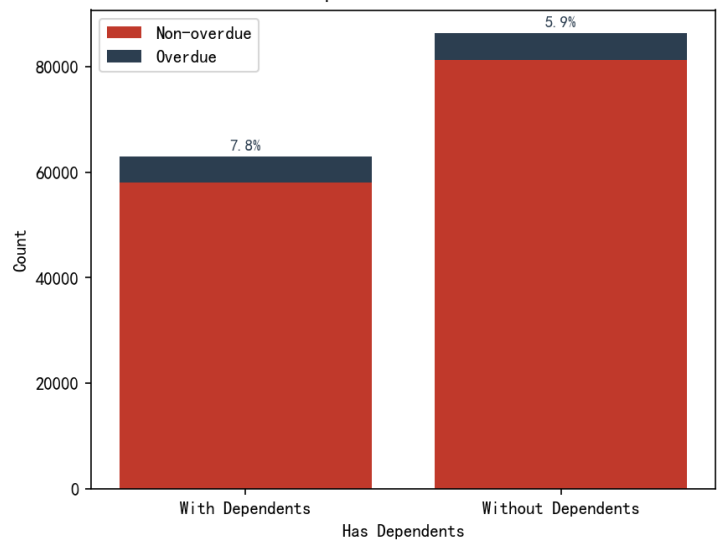
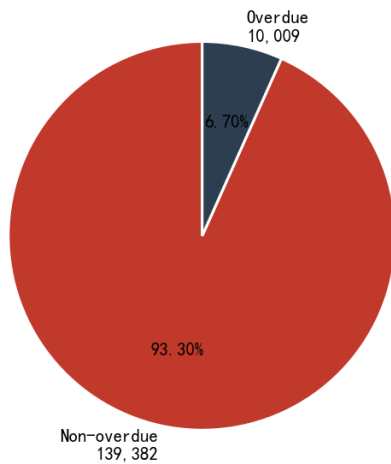


Figure 1.5. Number of dependents and overdue risk.

Monthly Income (Mean 6675) and Credit Card Overdue Analysis

Total Overdue Distribution by Monthly Income



Monthly Income and Overdue Status

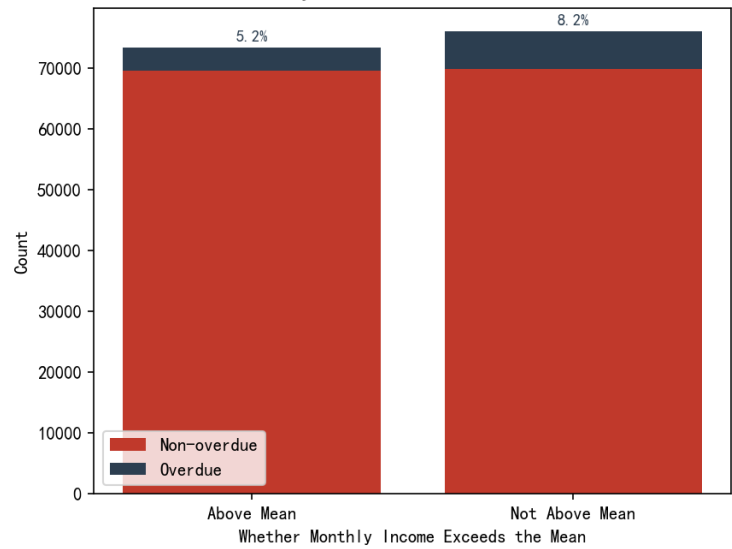


Figure 1.6. Monthly income and overdue risk.

The original dataset is imbalanced, and the most meaningful descriptive result is the large risk increase associated with historical overdue behavior. Age is also useful: younger customers show higher risk, while income, dependents, and real-estate-loan status have weaker associations.

3. Step 2 - Redoing the Analysis on the UCI Credit Card Default Dataset

The second part repeats the same analysis framework on a different dataset: UCI_Credit_Card.csv. The target variable is y, where 0 means non-default and 1 means default.

3.1 UCI Dataset Preprocessing

```
import pandas as pd
df = pd.read_csv("UCI_Credit_Card.csv")
df = df.rename(columns={"default.payment.next.month": "y"})
if "ID" in df.columns:
    df = df.drop(columns=["ID"])
df.to_csv("uci_data_clean.csv", index=False)
```

Item	Description
Number of records	30,000
Number of columns	24 after dropping ID
Target variable	y: 0 = normal / non-default, 1 = default
Key feature groups	LIMIT_BAL, demographics, repayment status PAY_0 to PAY_6, bill amounts, and payment amounts

3.2 Main Statistical Results

Factor	Observed result
Overall default rate	22.12% (6,636 default cases out of 30,000 records)
Age	Default rates are higher for the youngest and oldest groups; age 25 or below has 26.66% default rate.
Gender	Male customers have a higher default rate (24.17%) than female customers (20.78%).
Education	High-school customers have the highest default rate among major education groups (25.16%).
Marriage	Married customers show a higher default rate (23.47%) than single customers (20.93%).
Credit limit	Default rate decreases from 31.79% in the low-limit group to 14.29% in the high-limit group.
PAY_0 repayment status	Default risk rises from 13.83% for on-time customers to 75.78% for 3-month delay.

3.3 Visual Results and Interpretation

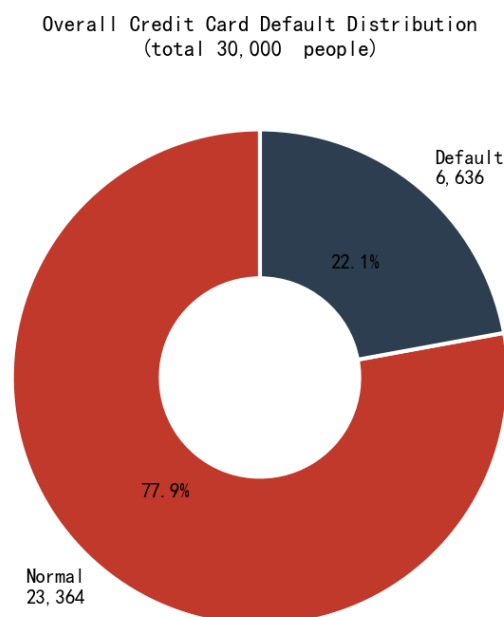


Figure 2.1. Overall default distribution in the UCI dataset.

Demographic Characteristics and Credit Card Default Analysis

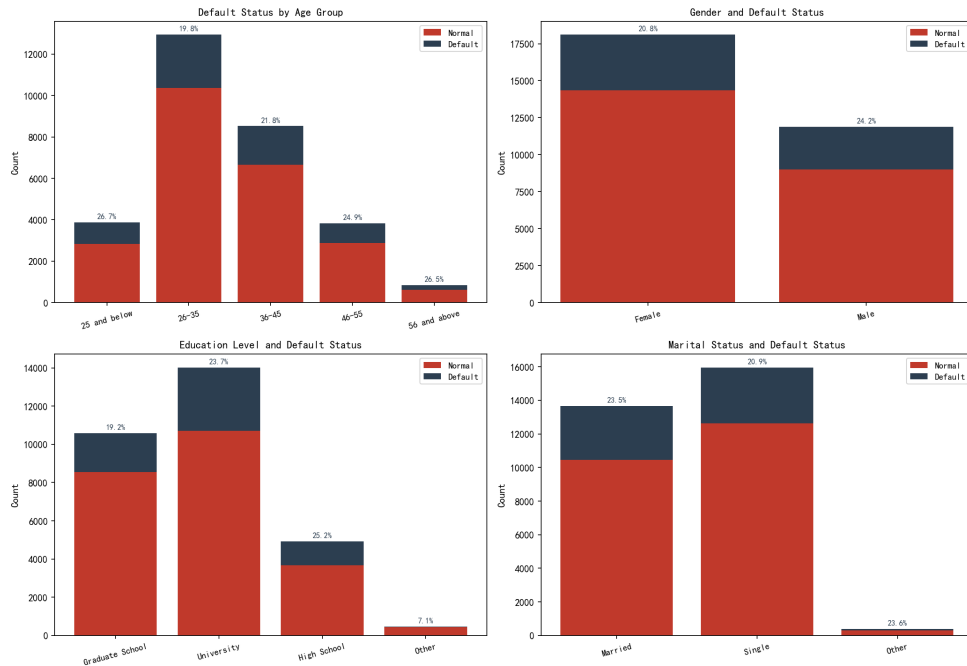


Figure 2.2. Default comparison by demographic variables.

Credit Limit (LIMIT_BAL) and Default Analysis

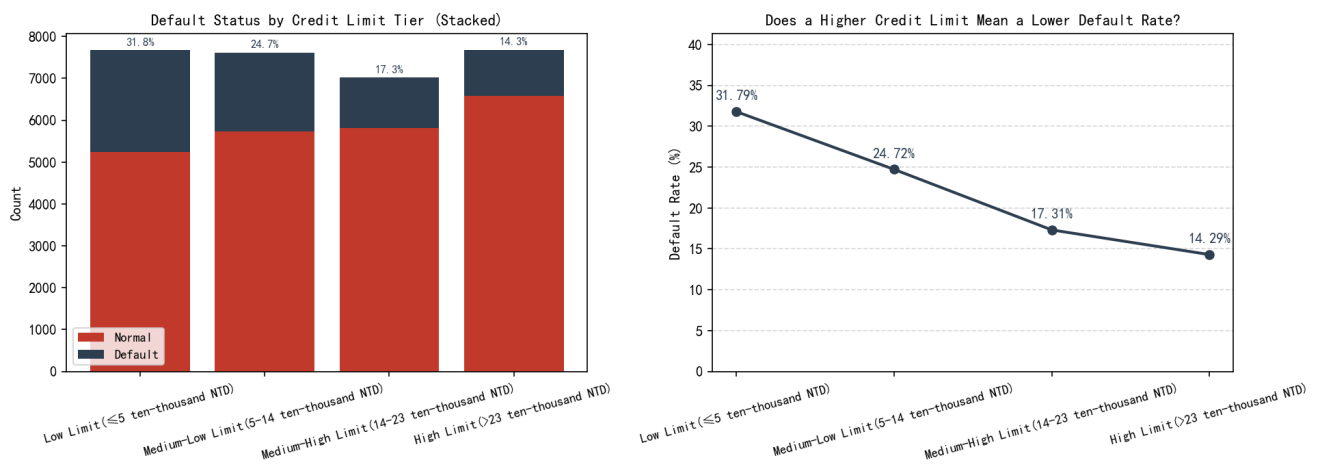


Figure 2.3. Credit limit group and default rate.

Repayment Delay Status (PAY_0) and Default Analysis

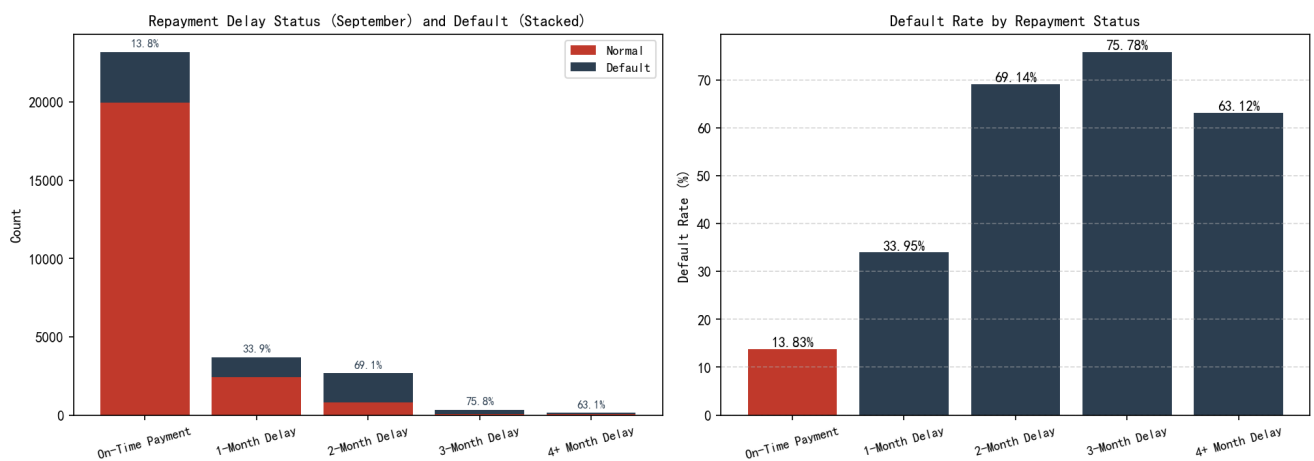


Figure 2.4. Repayment delay status PAY_0 and default rate.

Bill Amount and Payment Amount Trends (Normal vs Default)

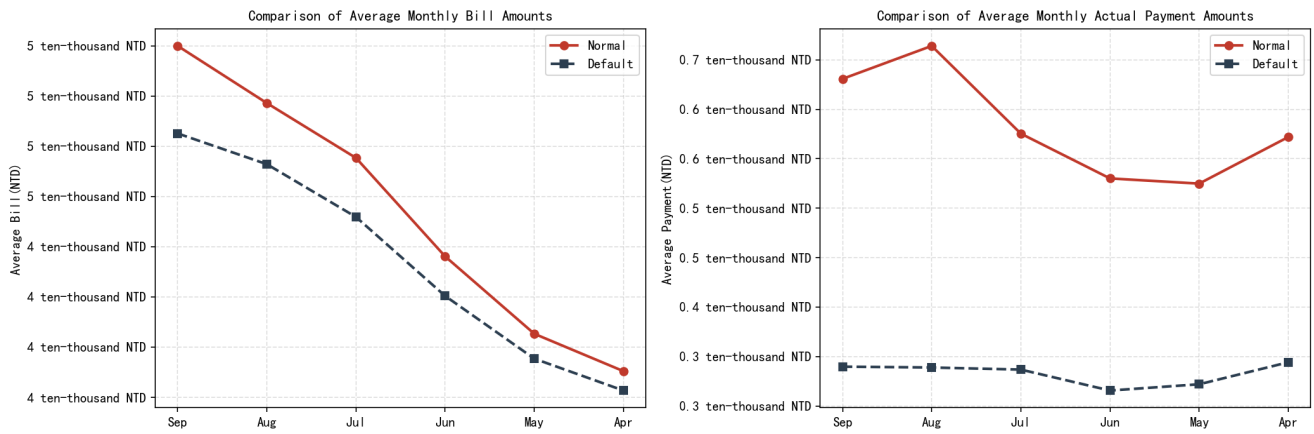


Figure 2.5. Bill amount and repayment amount trends.

The UCI dataset confirms the same core credit-risk pattern: repayment status and credit limit are more informative than most demographic variables. PAY_0 is especially important because it directly describes recent repayment behavior.

4. Step 3 - Machine Learning Extension

The machine learning extension uses the UCI dataset to predict customer default and to build customer risk segments. The target is y, where 1 means default.

4.1 Classification Task Design

```
FEATURE_COLS = [
    "LIMIT_BAL", "SEX", "EDUCATION", "MARRIAGE", "AGE",
    "PAY_0", "PAY_2", "PAY_3", "PAY_4", "PAY_5", "PAY_6",
    "BILL_AMT1", "BILL_AMT2", "BILL_AMT3", "BILL_AMT4", "BILL_AMT5", "BILL_AMT6",
    "PAY_AMT1", "PAY_AMT2", "PAY_AMT3", "PAY_AMT4", "PAY_AMT5", "PAY_AMT6"
]
assembler = VectorAssembler(inputCols=FEATURE_COLS, outputCol="raw_features")
sdf_ml = assembler.transform(sdf).select("raw_features", "y")
train_data, test_data = sdf_ml.randomSplit([0.8, 0.2], seed=42)
```

Item	Result
Training set	24,032 records
Test set	5,968 records
Training non-default customers	18,715 (77.88%)
Training default customers	5,317 (22.12%)

4.2 Model Training and Evaluation

Model	AUC-ROC	Accuracy	Precision	Recall	F1
Logistic Regression	0.7057	0.8043	0.7844	0.8043	0.7619
Random Forest	0.7710	0.8130	0.7932	0.8130	0.7897

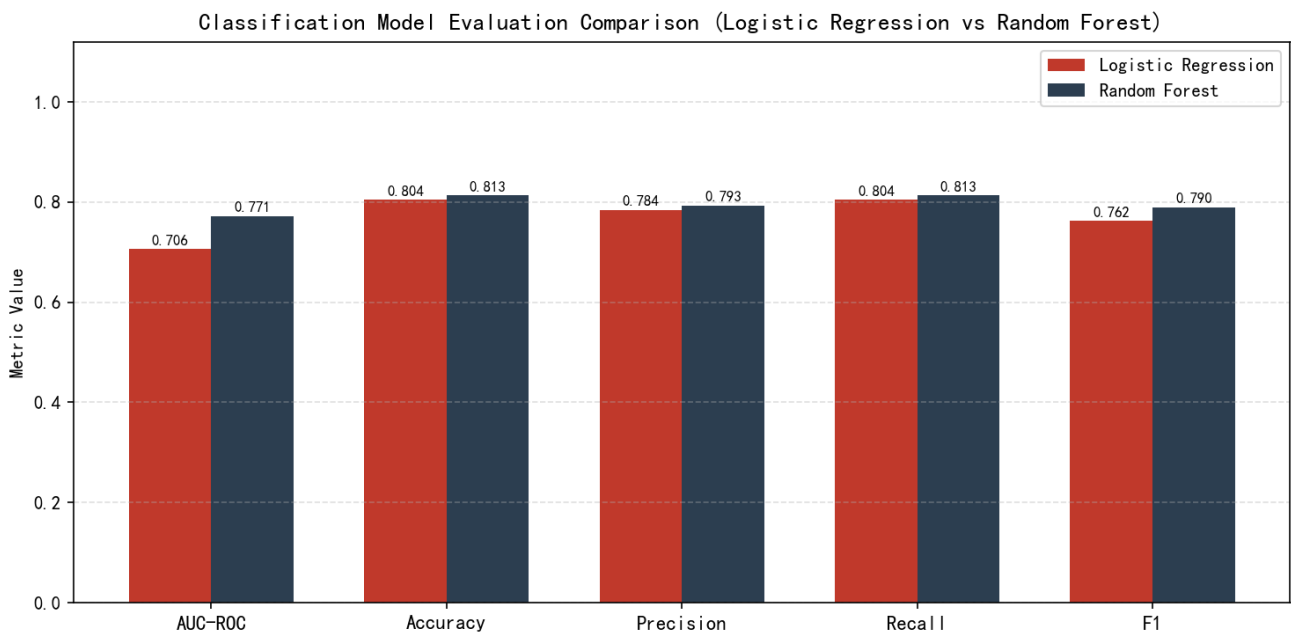


Figure 3.1. Classification model comparison.

Random Forest performs better than Logistic Regression on all major metrics. Its AUC-ROC of 0.7710 indicates better separation of default and non-default customers.

4.3 Feature Importance

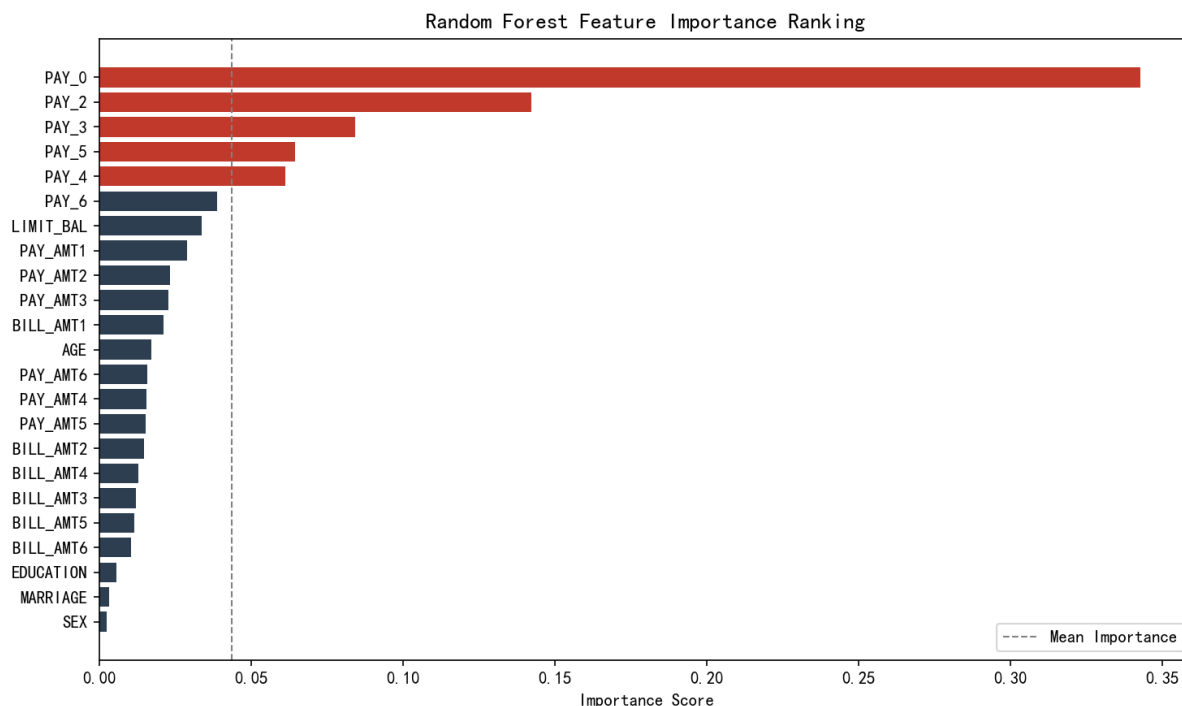


Figure 3.2. Random Forest feature importance.

Rank	Feature	Importance	Interpretation
1	PAY_0	0.3428	Most recent repayment status; strongest risk signal.
2	PAY_2	0.1421	Second recent repayment status.
3	PAY_3	0.0842	Historical repayment status.
4	PAY_5	0.0644	Historical repayment status.
5	PAY_4	0.0613	Historical repayment status.

The feature importance result supports the descriptive analysis: recent repayment behavior matters more than demographic information. This should be interpreted as model-based importance rather than causality.

4.4 Confusion Matrix and Business Risk

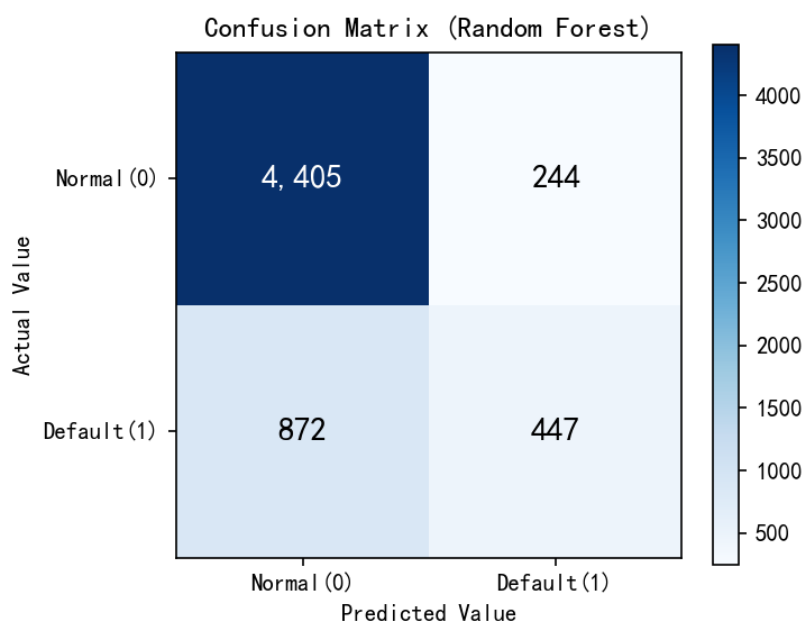


Figure 3.3. Random Forest confusion matrix.

Actual / Predicted	Predicted normal	Predicted default
Actual normal	4,405	244
Actual default	872	447

Although Random Forest reaches 81.30% overall accuracy, its class-1 recall is only 33.89%. In a banking context, false negatives are costly because actual default customers are incorrectly classified as normal. The model is useful for analysis, but it should not be used directly as a strict production risk-control model without threshold tuning, class weighting, or resampling.

4.5 K-Means Customer Segmentation

```
CLUSTER_COLS = ["LIMIT_BAL", "AGE", "PAY_0", "BILL_AMT1", "PAY_AMT1", "EDUCATION"]
cluster_assembler = VectorAssembler(inputCols=CLUSTER_COLS, outputCol="cluster_raw")
scaler = StandardScaler(inputCol="cluster_raw", outputCol="features_c", withMean=True, withStd=True)
```

```
for k in range(2, 9):
    kmeans = KMeans(featuresCol="features_c", k=k, seed=42)
    model = kmeans.fit(prepped)
    preds = model.transform(prepped)
```

K-Means Hyperparameter Selection

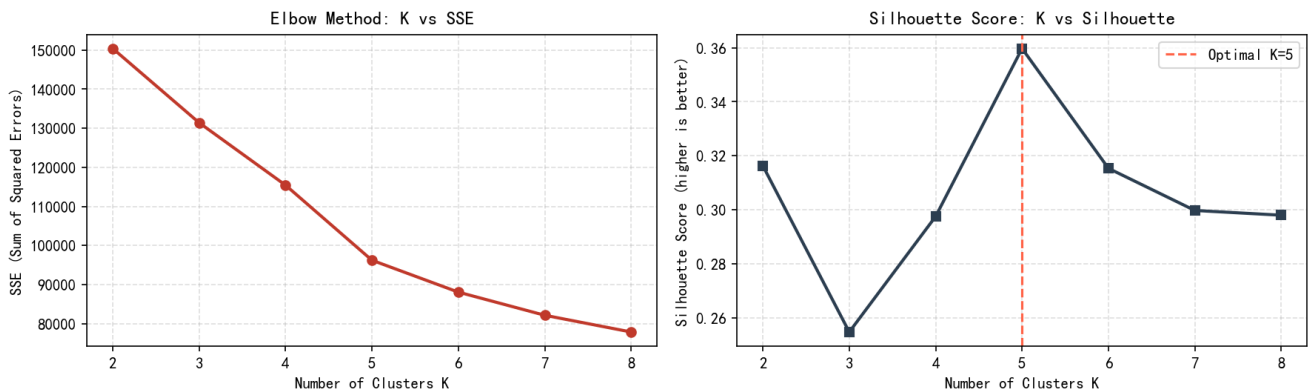


Figure 3.4. K-Means selection using SSE and silhouette score.

K-Means Clustering Results (K=5)

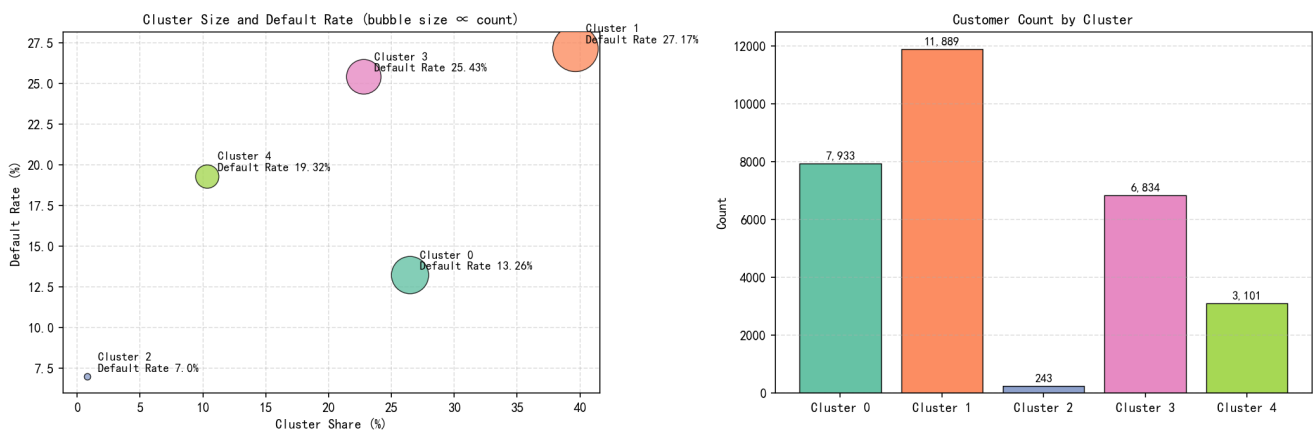


Figure 3.5. Cluster size and default rate overview.

Cluster	Customers	Share	Default rate	Interpretation
0	7,933	26.44%	13.26%	High credit limit, good repayment status, low risk.
1	11,889	39.63%	27.17%	Young, low credit limit, highest-risk group.
2	243	0.81%	7.00%	Special small group with high repayment amount and low risk.
3	6,834	22.78%	25.43%	Older, lower credit limit, high-risk group.
4	3,101	10.34%	19.32%	High credit limit, high bill amount, medium risk.

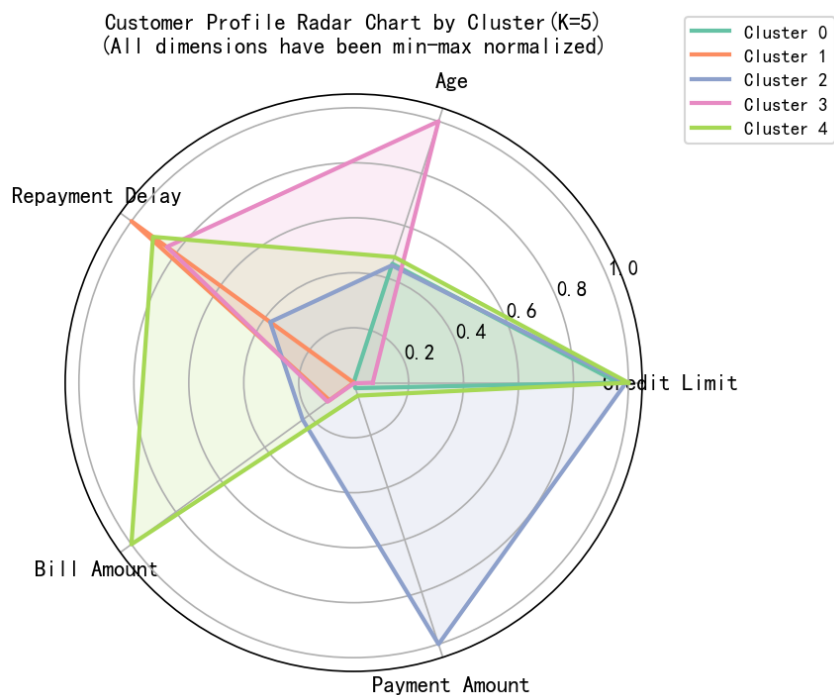


Figure 3.6. Cluster radar profile.

K-Means complements classification by grouping customers into interpretable risk profiles. Cluster 1 and Cluster 3 combine meaningful customer volume with high default rates and should receive more attention in risk management.

5. Integrated Findings

- Repayment history is the strongest signal across both descriptive statistics and model feature importance.
- Credit limit is negatively related to default risk in the UCI dataset.
- Demographic variables add context but are weaker than repayment behavior.
- The Random Forest model improves AUC, but the low default recall reveals a practical limitation.
- Customer segmentation makes risk easier to explain and supports differentiated monitoring strategies.

5.1 Limitations and Future Improvements

- The datasets are imbalanced, so accuracy can hide poor minority-class detection.
- The model threshold should be tuned according to the business cost of false negatives and false positives.
- Class weighting, oversampling, undersampling, Gradient Boosted Trees, XGBoost, or LightGBM could be tested in future work.
- PR-AUC and default-class recall should be emphasized in addition to AUC-ROC and accuracy.

5.2 Final Conclusion

The updated credit report shows that repayment behavior, especially recent repayment delay, is the central risk signal. Random Forest provides the strongest tested classification performance, while K-Means identifies actionable customer risk segments. The most important caution is that high overall accuracy does not guarantee strong detection of true default customers.

Teaching Evaluation			
	Course Section List	Instructor	Status
1	Experiential Arts (Ancient Court Music I) (1006)	Dr. Chia-Ying YEH	Evaluate
2	English for Academic Purposes III (1051)	Dr. John Chon Io WONG	Evaluate
3	Data Processing Workshop II (1002)	Dr. Yuchen ZHANG	Evaluated
4	Regression Analysis (1002)	Dr. Zhijian LI	Evaluate
5	Database Management Systems (1002)	Dr. Rui MENG	Evaluate
6	Chinese Culture and Modern China (1020)	Ms. Michelle Wenqian WANG	Evaluate

Teaching Evaluation			
	Course Section List	Instructor	Status
1	Tennis (1003)	Dr. Eric Derui TANG	Evaluate
2	Experiential Arts (Film Arts) (1022)	Ms. Mandy Man LI	Evaluate
3	English for Academic Purposes III (1051)	Dr. John Chon Io WONG	Evaluate
4	Data Processing Workshop II (1002)	Dr. Yuchen ZHANG	Evaluated
5	Regression Analysis (1002)	Dr. Zhijian LI	Evaluate
6	Database Management Systems (1002)	Dr. Rui MENG	Evaluate
7	Chinese Culture and Modern China (1019)	Ms. Jenny Mengya ZHANG	Evaluate

Teaching Evaluation			
	Course Section List	Instructor	Status
1	Experiential Arts (Chinese Ink painting) (1035)	Mr. Alen Jinlin XIE	Evaluate
2	English for Academic Purposes III (1055)	Dr. Guozhi CAI	Evaluate
3	Database Management Systems (1002)	Dr. Rui MENG	Evaluate
4	Data Processing Workshop II (1002)	Dr. Yuchen ZHANG	Evaluated
5	Regression Analysis (1002)	Dr. Zhijian LI	Evaluate
6	Chinese Culture and Modern China (1020)	Ms. Michelle Wenqian WANG	Evaluate